

The access of microfinance institutions to financing via the worldwide crowd

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Highlights

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We study the characteristics of microfinance institutions having access to refinancing via crowdfunding.

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We find evidence that the MFI's social performance is a main predictor of this kind of refinancing.

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We find evidence that the MFI's financial performance negatively related with this kind of refinancing.

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We examine those cases in which the partnership between the MFI and the platform has been terminated.

Abstract

The rising funding demand in the microfinance industry prompts microfinance institutions (MFIs) to seek additional, emergent debt capital sources such as crowdfunding. As the crowd-based approach has experienced widespread growth, we study the characteristics of MFIs having and using access to refinancing microloans via crowdfunding based on a panel data set obtained from the peer-to-peer microfinance platform Kiva. By performing binary regressions, we find evidence that the MFI's social performance regarding granting loans to women and the interest rate charged from borrowers are main predictors of refinancing through Kiva. We observe that mature MFIs exhibit better access to funding from Kiva than those which are new to the market. The results show that the likelihood of refinancing microloans through Kiva is negatively related to the financial performance

and to the extent of deposits of an MFI. MFIs operating in less-developed countries appear to be more likely to have access to Kiva's refinancing model. Additionally, we examine those cases in which the partnership between the MFI and Kiva has been terminated and find strong evidence that MFIs with a high share of deposits are more likely to discontinue the partnership with Kiva.

Introduction

Due to their rising funding demand, microfinance institutions (MFIs) tend to seek additional, emergent sources of debt capital such as crowdfunding. In the recent years, the crowd-based approach has experienced a widespread growth. This paper is the first to study the characteristics of those MFIs that have and also use access to refinancing their microloans via crowdfunding. Thereby the paper sheds light on the question under which conditions this refinancing source can play a role in the capital structure of an MFI.

Crowdlending as a new, crowd-based approach to fund loans to individuals and to small businesses, has emerged in the financial capital market since 2005 and is steadily growing regarding market size and importance. The concept of crowdlending implies that several individual investors enable the funding of a loan requested by another private person without the intermediation by a financial institution (Lin, Prabhala, & Viswanathan, 2013). The interest of individual investors to contribute with private capital to the debt market has been recognized by several commercial online peer-to-peer lending platforms (e.g. Prosper.com, Zopa, Auxmoney). Commercial online peer-to-peer lending platforms have been of academic interest in recent years (e.g. Emekter, Tu, Jirasakuldech, & Lu, 2015; Freedman & Jin, 2017; Hornuf & Cumming, 2017). However, this strand of literature is not directly related to our analysis, as we focus on crowdlending concepts with intermediaries and specifically analyze the relationship between the intermediary and the microfinancing platform.

In the microfinance industry, which aims at financing microentrepreneurs in developing countries, an increasing funding gap has become apparent in recent years as the funding demand of microfinance institutions exceeds the amount of donor capital, which remains the main source of debt capital

(Mersland & Urgeghe, 2013). A promising innovation that has the potential to diminish these credit constraints is crowdlending (Bruton, Khavul, & Siegel, 2015). Crowdlending has started to matter as source of debt capital provided by individuals as it is beyond charitable giving but much more a source of implicit subsidies in terms of low or no interest obligation against the investors (see Cull, Demirgüç-Kunt, & Morduch, 2018). Crowdlending offers the potential to be used as a source of debt capital by MFIs during their transition from the dependency on donor capital towards the ability to gain access to the mainstream capital market. The increasing relevance of crowdlending in the field of microfinance is reflected in a number of online microfinancing platforms (MYC4, Veecus, myELEN, Zidisha, unitedprosperity etc.). The most popular microfinancing platform is Kiva. Several studies on the decision making process and funding behavior as well as on the repayment behavior and credit risk provide valuable insights into the behavior of Kiva investors and microborrowers (Jenq, Pan, & Theseira, 2012; Meer & Rigbi, 2013). Besides Kiva investors and the microentrepreneurs at both ends, the core of this microfinancing model are the microfinance institutions (MFIs) which act as financial intermediaries. Thus, one cannot talk of (direct) peer-to-peer lending in this case but rather of an indirect form of lending from the crowd to the MFI. Kiva's microfinancing model is based on the financial intermediation by local MFIs which select and grant loans to microborrowers and monitor the repayment of the loans to Kiva investors. Kiva provides MFIs the possibility to refinance their loans without paying interest to the charitable investors. This paper is the first to analyze the characteristics of those MFIs that receive refinancing through international crowdfunding platforms. We thereby contribute to clarifying the question of how the leading microfinancing platform Kiva actually chooses MFIs to be funded as well as the question concerning what encourages MFIs to leave the partnership and abandon such a funding possibility.

Previous studies have found that microfinancing via Kiva is beyond charitable giving as the investors highly value the repayment of their capital. Therefore, Kiva itself and the MFIs have an incentive to ensure repayment and the MFI's positive financial and social reputation in order to appeal to potential investors and preserve the possibility of interest-free refinancing (Ly & Mason, 2012). On the one hand, it is the responsibility of Kiva to establish

requirements, to select and partner with MFIs and to connect the MFI's needs with the expectations of the social investors. On the other hand, the MFI itself has to be willing to adopt Kiva's requirements and to approach the investors' expectations.

We derive a set of hypotheses from theoretical considerations, taking into account both the supply and the demand side perspective. In our empirical strategy, we connect information derived from MIX Market with information available on Kiva's Application Programming Interface (API) in order to identify the determinants of having access to Kiva and making use of it. Note that the access notion we employ in the remainder of this paper captures the fact that Kiva is willing to provide access to refinancing and at the same time that the MFI makes use of it. Our panel data set includes several social and financial performance indicators. By performing binary regressions, we ascertain that the MFI's social performance in terms of granting loans to women and charging low interest rates is a main predictor of having and using access to Kiva. Several maturity variables appear to show a significant correlation with the probability of access, which leads us to the conclusion that mature, more experienced MFIs have access to funding from Kiva more frequently than new, less-experienced MFIs. Furthermore, we find evidence that MFIs with a solid operational self-sufficiency¹ and a large extent of deposits are less likely to have access to Kiva. MFIs operating in poorer countries appear to exceed their peers in high-GDP countries in terms of having access to funding through Kiva. Additionally, we examine those cases in which the partnership between Kiva and an MFI has been terminated. Based on the same influential variables, we perform binary regressions and Cox proportional hazard models. As the termination of the partnership is initiated² by the MFI, either directly or indirectly by intentionally violating certain requirements, this approach enables us to shed some light on demand side aspects. The share of female borrowers reveals itself to be not only a positive driver of access to Kiva, but also a predictor of the termination of the partnership. In contrast to the access results, neither the maturity, the portfolio quality, nor the operational self-sufficiency have predictive power. However, we find strong evidence that MFIs which are able to mobilize deposits as a sustainable, independent source of debt capital have a lower

tendency to retain the partnership with Kiva, which requests an MFI to fulfill certain (ethical and financial) requirements.

The remainder of this article is organized as follows: In Section 2, the Kiva microfinancing model is outlined. In Section 3, we develop a theory and derive some hypotheses regarding the influence of institutional determinants.

Section 4 describes the data and the regression models employed. Empirical findings regarding refinancing through Kiva and termination of the partnership as well as several robustness checks are reported in Section 5.

Section 6 concludes.

Section snippets

The Kiva microfinancing model

Kiva, a donation-based NGO, is an online crowdlending platform that facilitates microborrowing by connecting socially oriented investors with the poor aiming to support their small businesses. Until mid of 2018, microloans of 1.2 billion USD have been transferred into 85 countries through partnering with local MFIs (Kiva.org, 2018). The core of Kiva's microfinancing model is the partnership with local MFIs. The main actors in Kiva's microfinancing model are displayed in Fig. 1.

MFIs act as

Theory and hypotheses development

In this section, we first give a short summary of MFI funding, then develop a theory regarding the access to refinancing via the crowd and finally employ this theory to derive a set of hypotheses in a second step.

Data description

In our study, we combine MFI-specific financial and social performance indicators reported to the online platform of the Microfinance Information Exchange (MIX Market) with the information on the MFI's activity on Kiva as a source of subsidized debt funding. MFI-specific data, such as scale variables,

social performance indicators, and financial ratios, are derived from MIX Market. In order to maintain good data quality, only data classified with a minimum of a disclosure rating of 3 diamonds¹¹

Results on the access to debt capital from Kiva

In this section, we investigate the access to funding from Kiva as dependent variable. As already noted, access can only be observed if both Kiva and the MFI mutually agree upon the funding relationship. Table 6 exhibits the results of the estimated pooled logistic models. Model I includes the entire 5621 observations in the panel data set. Model I is extended by the squared debt-to-asset ratio, resulting in model II. Model III is the base model including the real portfolio yield as a further

Conclusion

We empirically study the access of MFIs to subsidized funding provided via the online peer-to-peer platform Kiva between 2005 and 2015 based on a worldwide dataset of 909 MFIs. By employing logit regressions, we identify a positive relationship between the MFI's social performance and access to funding from Kiva. Regarding the MFI's maturity, we find strong evidence that more mature MFIs in terms of debt-to-asset ratio have and use access to Kiva more frequently than MFIs which are new to the

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Citation Excerpt :

Research avenue 1 focuses on exploring the effects of individual antecedents on the actors' participation in different crowdfunding forms from the perspective of (a) entrepreneurs and (b) backers in sustainability-oriented crowdfunding. Extant research provided relevant findings regarding the motivational and behavioral mechanisms of individuals, especially backers, by examining campaign-related antecedents and outcomes on crowdfunding platforms (e.g., Dorfleitner et al., 2020; Gleasure and Feller, 2016). However, our review still reveals ambiguities and ongoing debates about the relevance of, for example, extrinsic and intrinsic motivations (e.g., Allison et al., 2015; Kollenda, 2022) or altruistic and warm-glow motivations (e.g., Allison et al., 2013; Berns et al., 2020), as well as the extent of rational behavior by crowdfunding actors (e.g., Bento et al., 2019; Yoo et al., 2023).

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In view of the results, Tsinghua University cooperating with the National University of Singapore and the University of Michigan to study crowdfunding (Ai et al., 2016) has the most extensive partnership. Apart from that, the University Regensburg and Centre for European Research in Microfinance (CERMi), as well as Copenhagen Business School and Stockholm School of Economics, also conducted collaborative research on crowdfunding projects (Dorfleitner et al., 2020; Nielsen, 2018). Additionally, some institutions such as North Dakota State University and Northern Arizona University worked

together to study P2P lending (Dorfleitner & Oswald, 2016; Riggins & Weber, 2015).

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